Selection on Observables II: Estimation

Stat 256, Spring 2026

Chad Hazlett

UCLA

chazlett@ucla.edu

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Today: *estimation*. Given a chosen adjustment set X, how do we actually compute $\widehat{\tau}$?

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These differ only in *how* they approximate conditioning on X.

Running example: LaLonde / Dehejia–Wahba

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Observational challenge. DIM on observational sample= $-\$15{,}205$. Eek! The PSID controls are nothing like NSW-treated: older, richer, more employed.

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Matching handles this by finding *near*-matches instead of exact ones.

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Within each cell, compute DIM; re-weight cells by the *treated* share (ATT):

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Why: Turns out the unemployed (in 1975) people in PSID and NSW are fairly comparable, and within just them you get reasonable estimates, but the employed in PSID and NSW are pretty different.

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- **Matching routine** — *how do we assemble the matched sets?*

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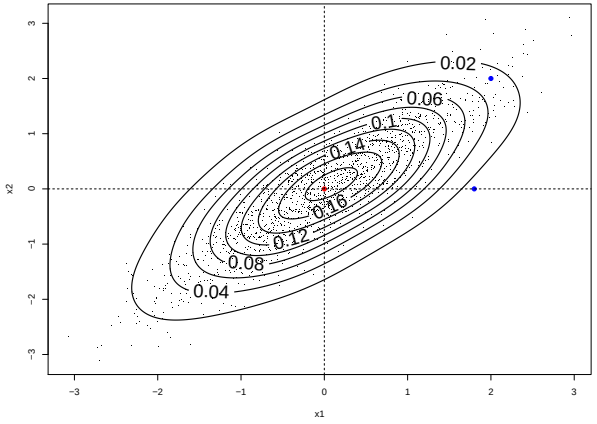

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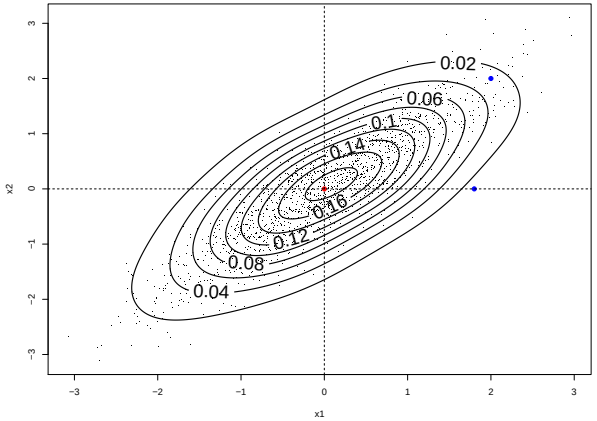

Others: coarsened exact matching, genetic matching, . . .

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Full / complete matching (Hansen 2004): every treated matched to ≥ 1 control *and* vice versa; variable-size sets. Uses all data.

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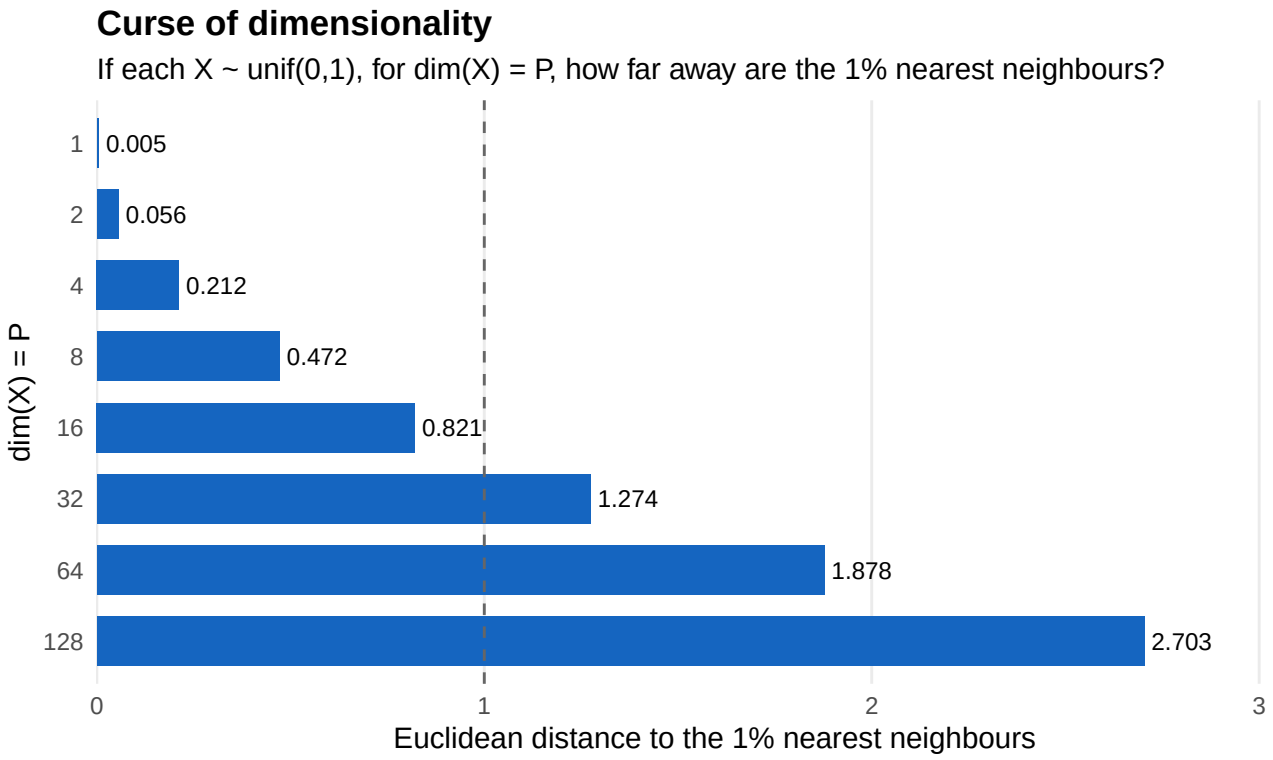
Bias-adjusted matching: fit $\widehat{\mu}_0(X)$ among controls, replace $Y_{j(i)}$ with $Y_{j(i)} - [\widehat{\mu}_0(X_{j(i)}) - \widehat{\mu}_0(X_i)]$. A "little regression" fills the gap.

Limits of matching (II): curse of dimensionality

As $\dim(X)$ grows, everyone is far away.



Broader point: Matching is appealingly nonparametric, but you pay a price in bias, especially as P grows.

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Costs:

- You have to specify a model for $\pi(X)$, if wrong, $\hat{\tau}$ inconsistent.
- You don't get "exact balance" on X in any sense.

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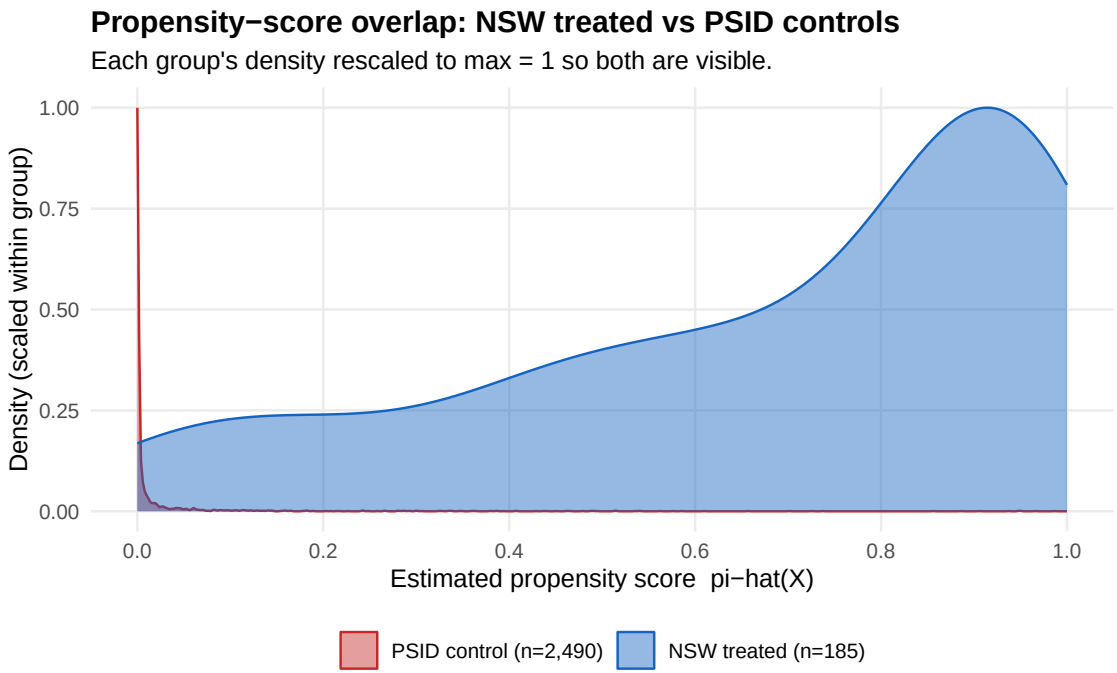

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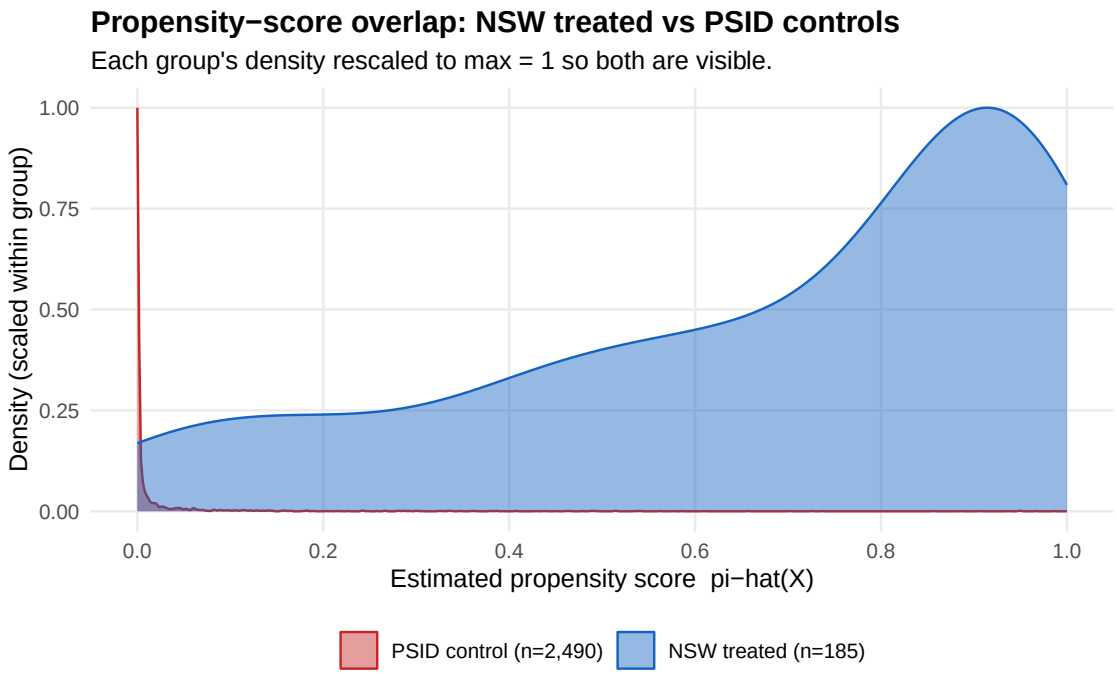

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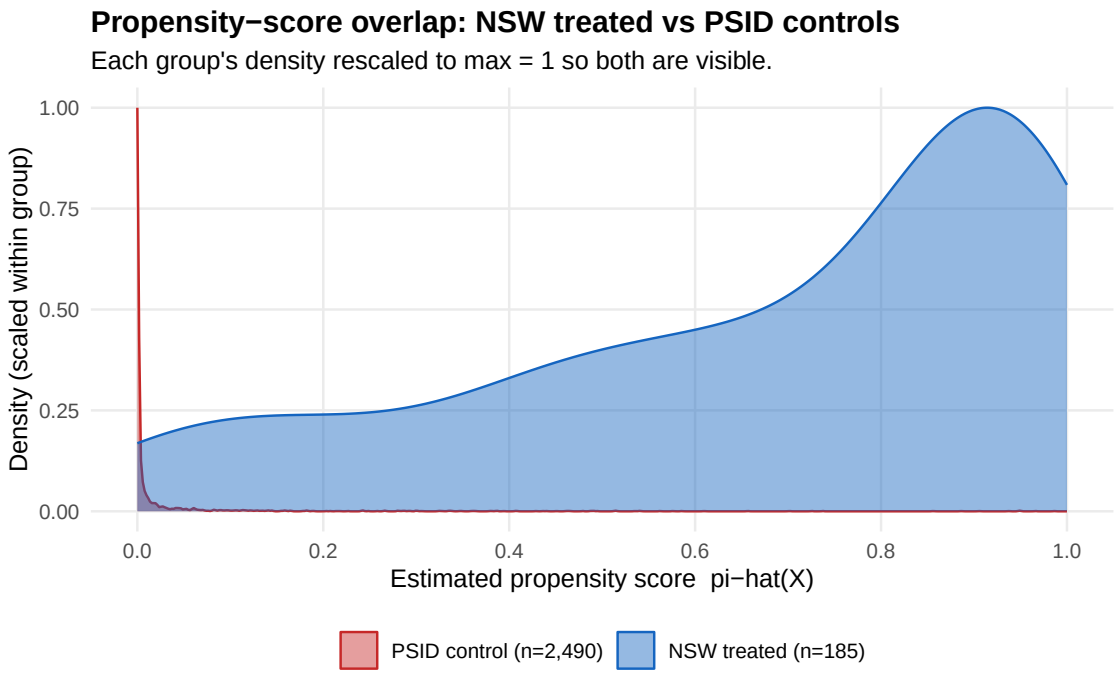

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If two-sided poor overlap, you wouldn't be able to do anything.

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These are the (**stabilised**) IPW weights:

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ATT version: treated $w_i = 1$ (already at target); controls $w_i = \dfrac{\widehat{\pi}(X_i)}{1 - \widehat{\pi}(X_i)} \cdot \dfrac{\Pr(D{=}0)}{\Pr(D{=}1)}$ (Bayes for $p(X \mid D{=}1)/p(X \mid D{=}0)$).

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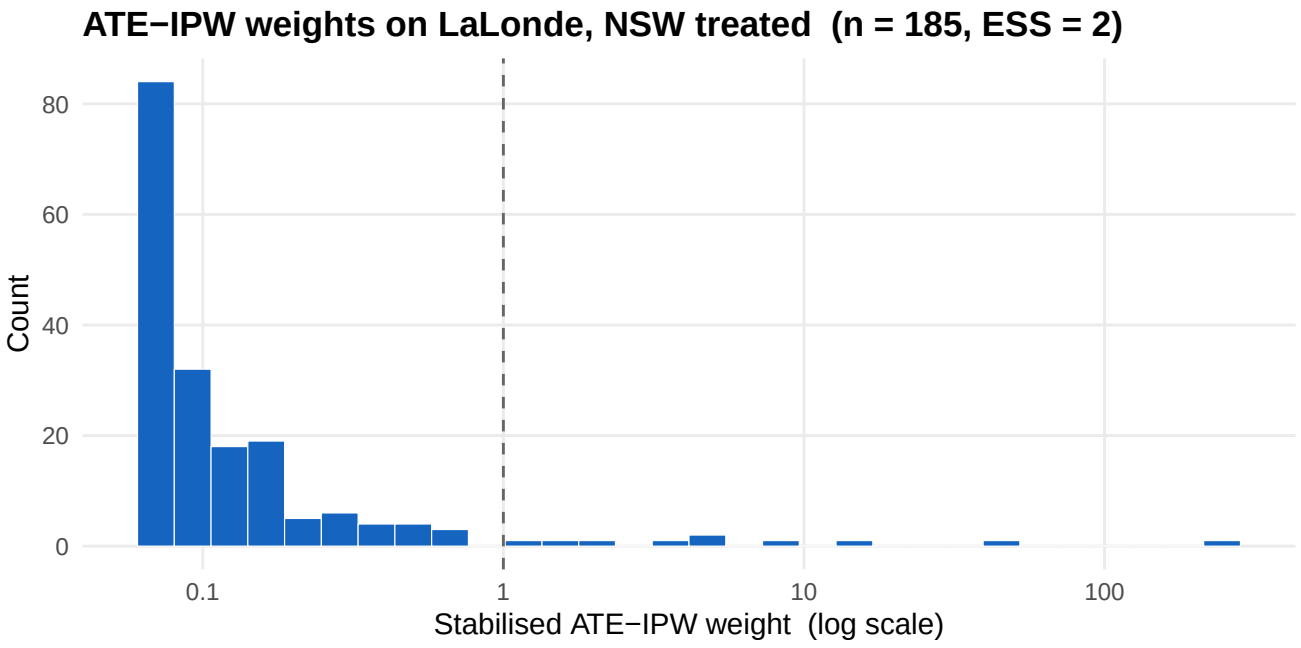

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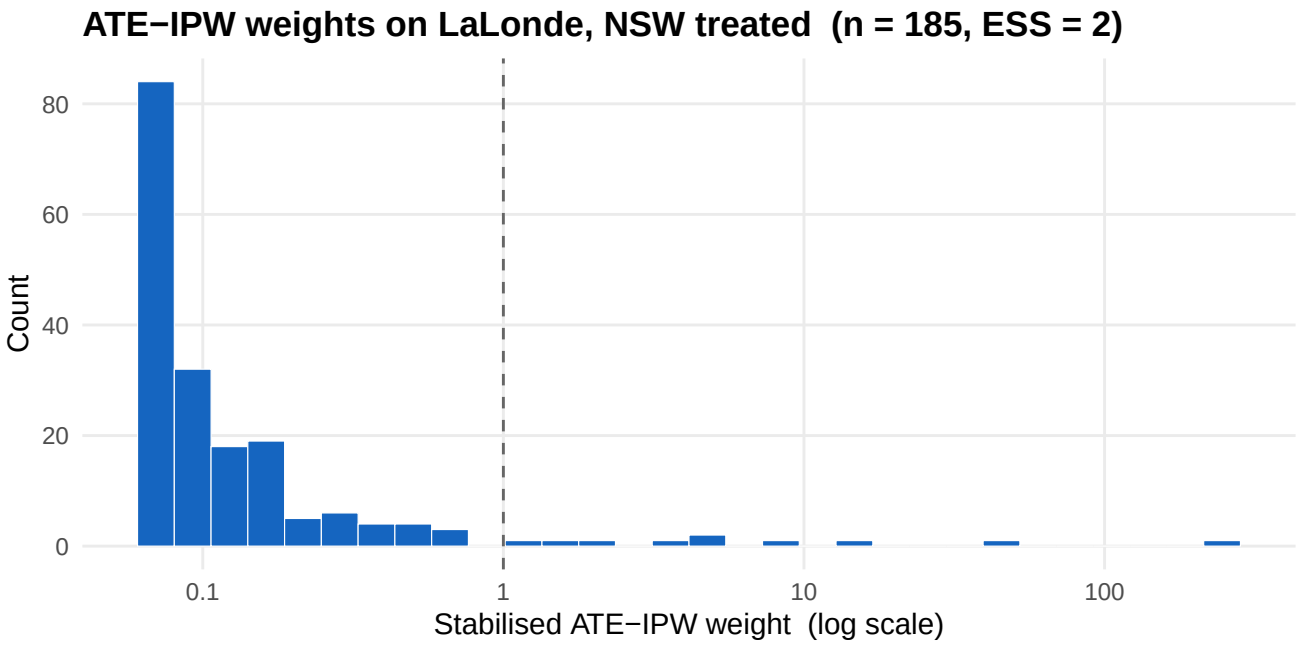

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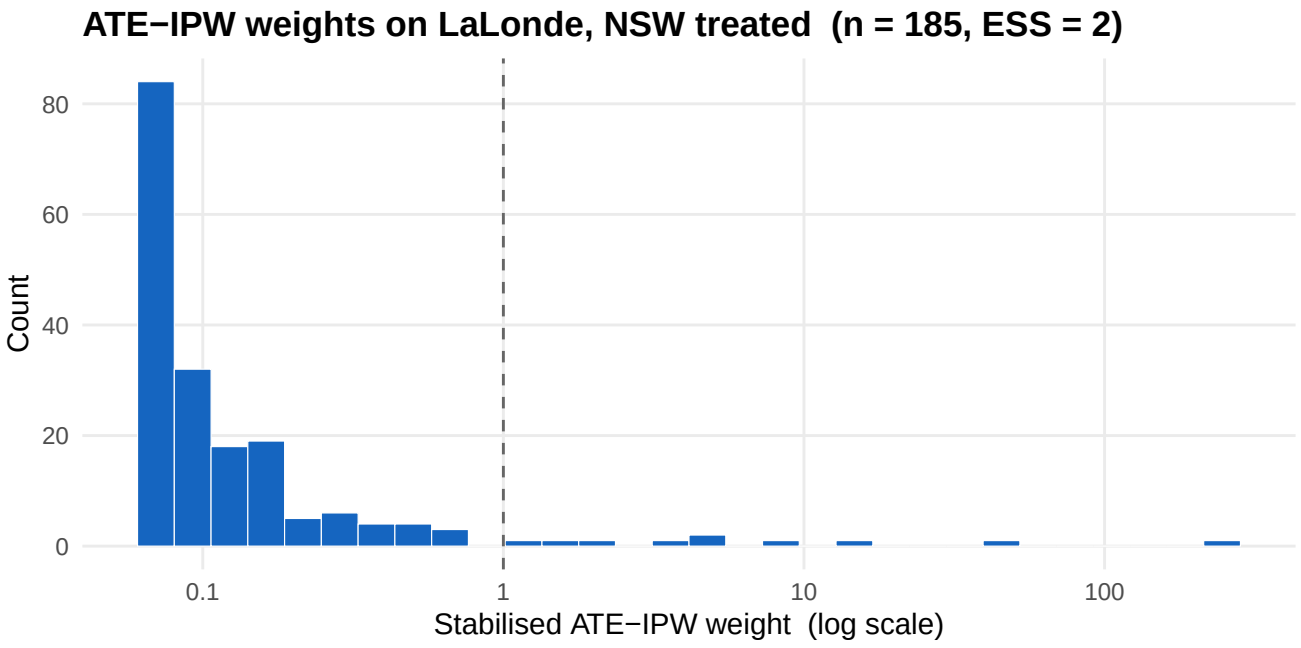

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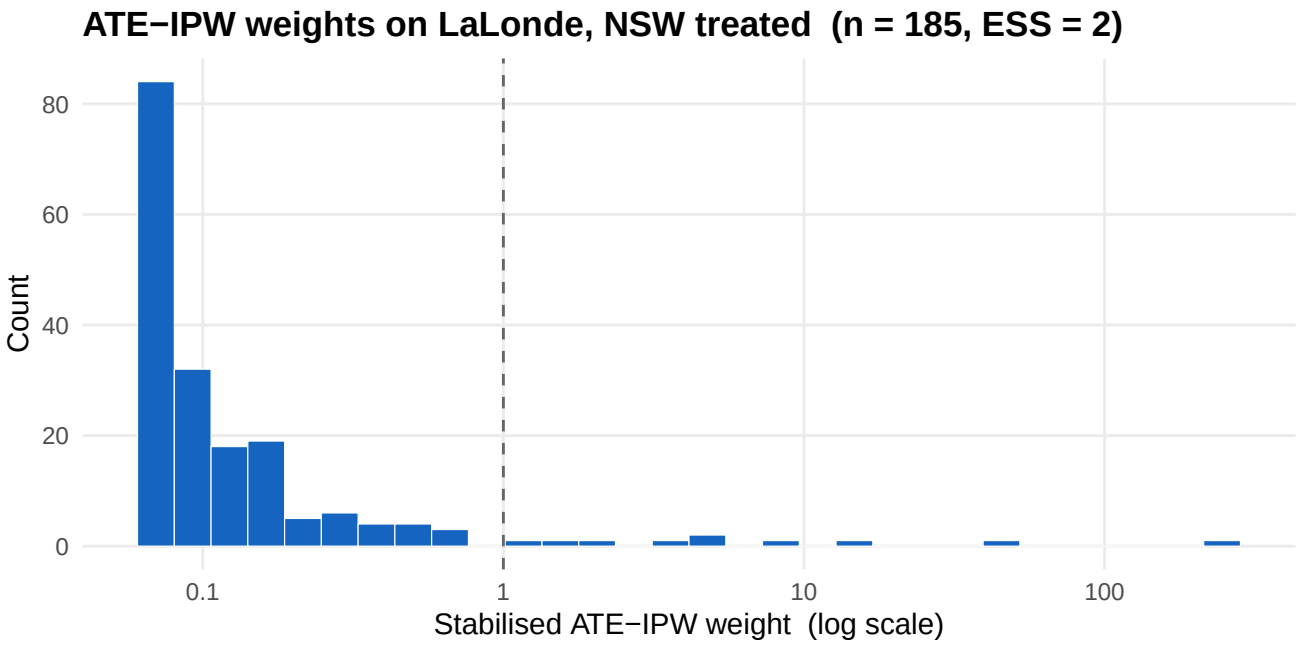

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- For treated, ESS $= (\sum w_i)^2 / \sum w_i^2 \approx 2$! (from 185).
- For controls, many get downweighted but ESS ≈ 711 (from 2490). ATT fine.

Balancing weights: solve directly

If you want $p(X \mid D = 1) \approx p(X \mid D = 0)$, why not just choose weights $\{w_i\}$ s.t. weighted control distribution matches treated (ATT) or target (ATE) on (chosen moments of) X?

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Mean-balance constraint (in ATT case):

$$\sum_{i:\, D_i=1} \phi(X_i) \;=\; \sum_{i:\, D_i=0} w_i\, \phi(X_i), \quad \text{s.t.} \sum w_i = n_1, \quad w_i \geq 0 \;\forall i$$

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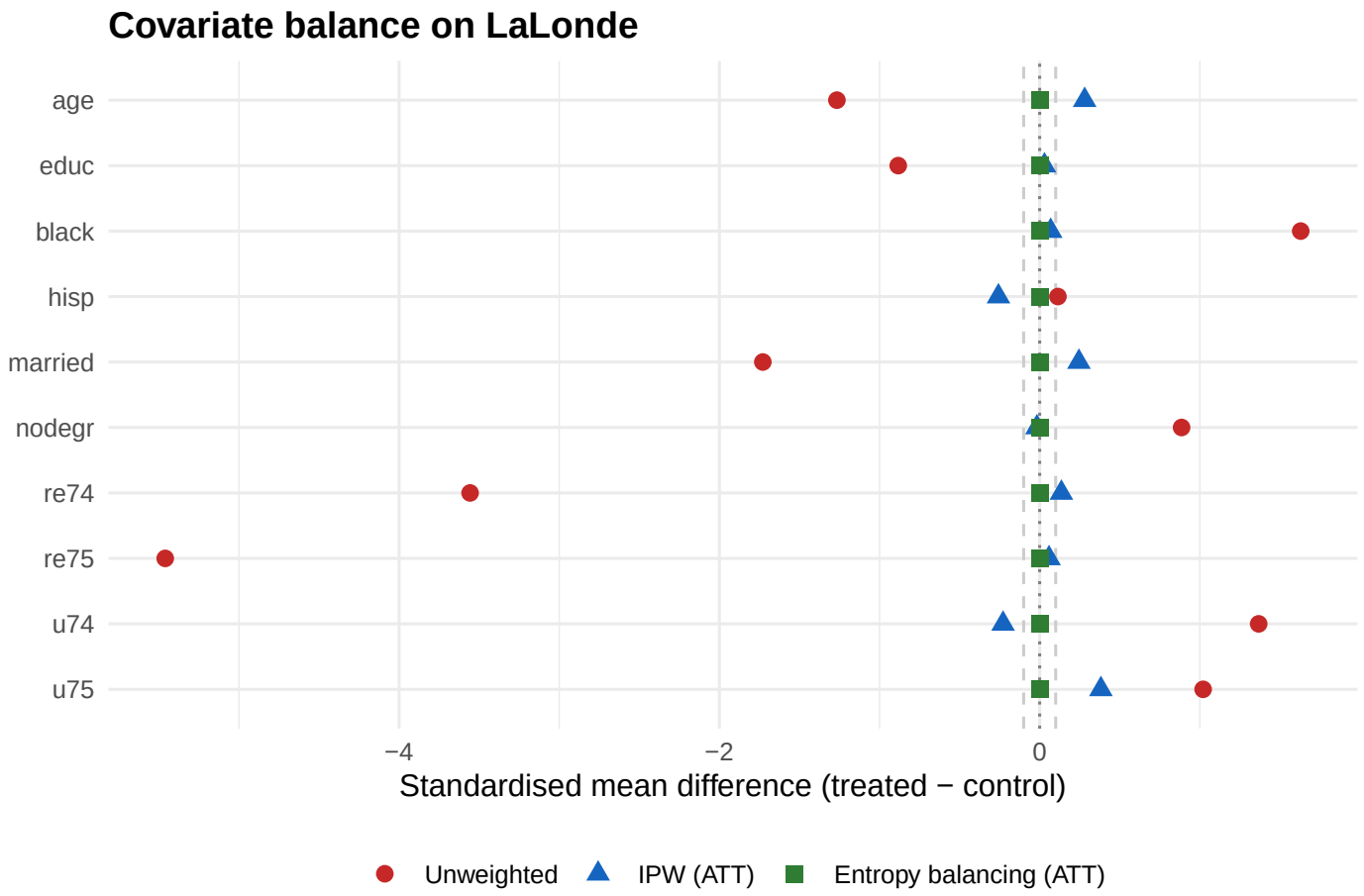
Constraint alone is under-determined. Add a regulariser on w such as:

- Maximum entropy: $\min \sum_i w_i \log w_i \rightarrow$ entropy balancing (Hainmueller 2012)
- L_2 penalty: $\min \sum_i w_i^2 \rightarrow$ stable balancing weights (Zubizarreta 2015)
- Empirical likelihood, $\sum_i log(w_i)$

Balance on LaLonde: unweighted vs IPW vs ebal

Standardised mean differences across covariates: unweighted is far off; IPW shrinks the gap but *doesn't* hit zero (and can still miss); entropy balancing zeroes the means by construction.



CBPS (Imai & Ratkovic 2014): a pscore tuned for balance

What if we still want a pscore (interpretable, parametric, easy to inspect) but tune it so the resulting IPW gives exact finite-sample balance? Stack the moment conditions per unit:

$$\phi(X_i, D_i; \beta) \;=\; \begin{pmatrix} \left[D_i - \pi(X_i;\beta)\right] X_i \\ \left[\dfrac{D_i}{\pi(X_i;\beta)} - \dfrac{1 - D_i}{1 - \pi(X_i;\beta)}\right] X_i \end{pmatrix}.$$

Top: standard logistic MLE score (what logistic regression solves). Bottom: IPW balance moment — forces IPW-reweighted means of X to match across arms.

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Choose β to set the moments to zero:

- **M-estimator** (just-identified): keep k of the $2k$ moments — e.g. balance only — and solve $\sum_i \phi^{(j)}(X_i;\beta) = 0$.
- **GMM** (over-identified, both rows): $\widehat{\beta} = \arg\min_\beta \left[\frac{1}{n}\sum_i \phi\right]' W \left[\frac{1}{n}\sum_i \phi\right]$.

Beyond means; prognostic balancing

What does mean balance buy? Choosing $\phi(X) = X$ balances X's means across arms — which translates into balance on *linear functions* of X.

- If $\mu_0(X)$ is linear in X, that's exactly what you need: $\mathbb{E}[\mu_0(X)]$ matches across arms and $\widehat{\tau}$ is unbiased.
- If $\mu_0(X)$ has nonlinearities or interactions, mean-balanced X does <u>not</u> imply balanced $\mu_0(X)$. Residual bias remains.

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- Fit $\widehat{\mu}_0(X)$ on controls; project onto all units.
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Efficient; handy especially when poor overlap on (non-prognostic) X. Work in progress.

G-computation / imputation

Instead of weighting, *model* $\mu_1(x)$ and $\mu_0(x)$. Then impute each unit's missing potential outcome:

$$\widehat{\tau} = \frac{1}{n}\sum_{i=1}^{n}[\widehat{\mu}_1(X_i) - \widehat{\mu}_0(X_i)]\,.$$

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Functional form of $\widehat{\mu}_d$ is your choice: OLS, random forest, kernel regression, ...

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Lin (2013). Fit the fully interacted model

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Two things to want from this regression.

- The coefficient on D should be the *average* treatment effect, not "effect at $X = 0$."
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We get both if we center X first.

Centering X: why and what it buys

Let $\tilde{X}_i = X_i - \bar{X}$ (sample-mean-centred). The interacted model becomes

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(i) Average effect.

$$\frac{1}{n} \sum_{i=1}^{n} \partial Y_i / \partial D \;=\; \tau + \gamma' \underbrace{\tfrac{1}{n} \sum_{i} (X_i - \bar{X})}_{=0} \;=\; \tau.$$

And the (linear) imputation estimator does just that:

Fitted means:

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So with centered X, three things collapse to one number:

- the coefficient $\widehat{\tau}$ on D,
- the average unit-level slope,
- the g-computation / T-learner imputation estimate.

All roads lead to the same place

Under CI, every method is estimating $\mathbb{E}_{X \sim F}[\mu_1(X) - \mu_0(X)]$.

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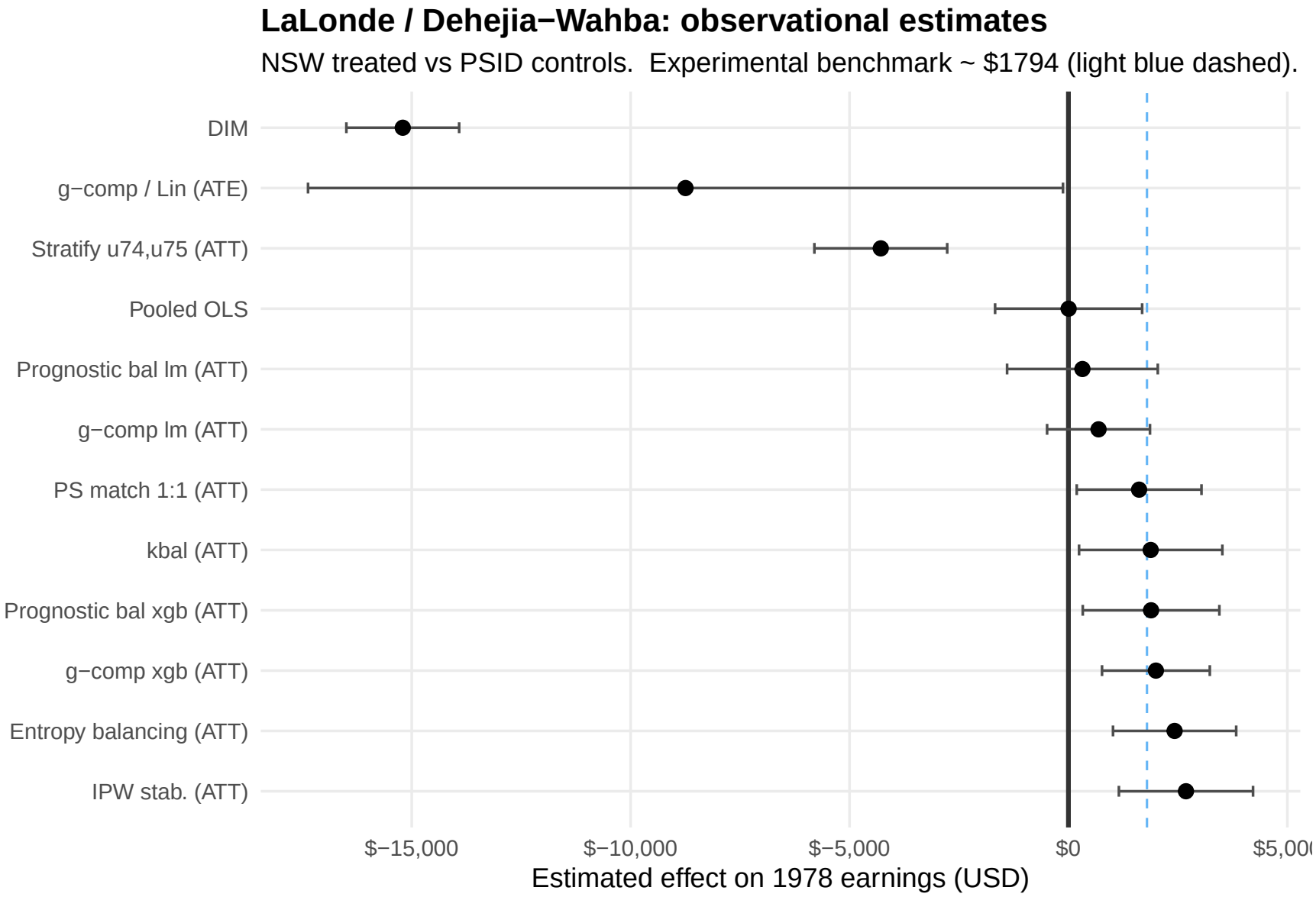
One angle: Choose the appropriate set of these for your problem, and implement them all. If they disagree, diagnose.

LaLonde: tough one due to overlap



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Augmented / doubly robust: a signpost

Methods that use **both** $\widehat{\pi}$ and $\widehat{\mu}_d$:

- **AIPW**: IPW estimator plus a regression adjustment (equivalently, g-comp plus IPW-weighted residuals). Consistent if *either* $\widehat{\pi}$ or $\widehat{\mu}_d$ is right.
- **TMLE**: iteratively tilt $\widehat{\mu}_d$ using $\widehat{\pi}$ to solve an efficient score equation.
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If nobody picks it, I might carve out a lecture.

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Extrapolation is not reweighting. On LaLonde, OLS-ATE extrapolates $\widehat{\mu}_1$ far outside the treated support and lands at $-\$8{,}746$; the ATT version is $+\$688$; weighting methods cluster near the truth. *Pick estimand and method with an eye on support.*

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Method menu. Subclass / matching / PS-matching / IPW / balancing weights / regression imputation (Lin). Pick based on what you're willing to model.

Lin: interacted OLS on *centered* X gives a D-coefficient equal to the g-computation estimate. Center at the right mean for your estimand.

Extrapolation is not reweighting. On LaLonde, OLS-ATE extrapolates $\widehat{\mu}_1$ far outside the treated support and lands at $-\$8{,}746$; the ATT version is $+\$688$; weighting methods cluster near the truth. *Pick estimand and method with an eye on support.*

What's still missing. Unobserved confounding. When CI fails, every method on this menu is biased.